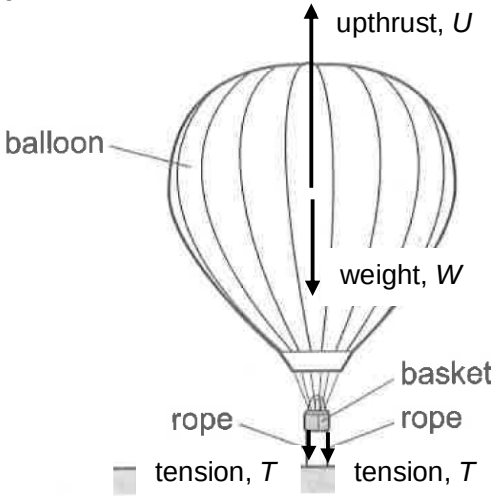


8	A	Weight of basket and uninflated balloon $= 700 \times 9.81$ $= 6867 \text{ N}$ Weight of hot air in inflated balloon $= 0.898 \times 2800 \times 9.81$ $= 24666 \text{ N}$ Upthrust $= 1.204 \times 2800 \times 9.81$ $= 33071 \text{ N}$ Since hot air balloon is in equilibrium tension in each rope $= \frac{1}{2}(33701 - 24666 - 6867)$ $= 769 \text{ N}$ 
8	D	Useful power output, P